What Can Be Computed

Ten examples on our data:
from the atom to the whole

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The Chain

The model computes with few inputs and few formulas. The inputs are the constants of nature (c, G, ħ, the fine-structure constant α), the mass of the body and a single threshold, $a_0 = 1.06 \cdot 10^{-10}$ m/s^2. For the whole, three measured values are added (Planck): the baryons ω_b, the radiation via T_{CMB}, and ω_c. ω_c is not a substance of its own but the weight of the return in the whole, $Q = \omega_c/\omega_b$. Everything else follows.

What	Formula	What it gives
Fold	$\Delta s = GM/(3c^2)$	the measure of mass: the extra length at the centre of the funnel
Slope	$g = 3c^2\Delta s/r^2$	the acceleration of fall, Newton
Pace	$d\tau/dt = \sqrt{(1 - 6\Delta s/r)}$	the rate of clocks in the funnel
Return	$a = a_N/(1 - q)$, $q = e^{-\sqrt{(a_N/a_0)}}$	motion at every acceleration: Newton above a_0, flat rotation curves below
Threshold radius	$r^* = \sqrt{(GM/a_0)}$	where a body reaches the threshold
Breathing	$\rho_{DE} = 3\pi^3 a_0^2/(30Gc^2)$	dark energy and with it H_0 and Ω_{DE}
Orbit of the shell	$v = \alpha c$ on a_B	the binding of the atom
Holder of the nucleus	$a_C = (3/5)\cdot\alpha\hbar c/\lambda_H$	the length on which the nucleus holds its nucleons

The **return fraction** q is the part of a motion that returns through the inner direction. a_N is the acceleration that Newton alone would give from the visible mass. Above the threshold q is nearly zero and the rule is Newton. Below it q grows and holds the orbits (Figure 1). The threshold is the same everywhere: galaxies, the distances of supernovae and the breathing of the whole read the same number.

What can be computed cleanly in this way:

- **Particles and nucleus:** the binding of hydrogen, the holder length of the nucleus from AME2020, the closure numbers of shells and nuclei, the end of matter at Z = 118.
- **Stars and galaxies:** the rotation curve from the visible alone, the threshold from 141 galaxies (SPARC), the disc length $R_d = r^*/\sqrt{2}$.
- **The whole:** $H_0 = 67.2$ km/s/Mpc and $\Omega_{DE} = 0.685$ from a_0 and the three measured values of the whole, the shape of the expansion from supernovae (Pantheon+) and BAO (DESI DR2).

Ten examples follow, step by step, from the atom to the whole.

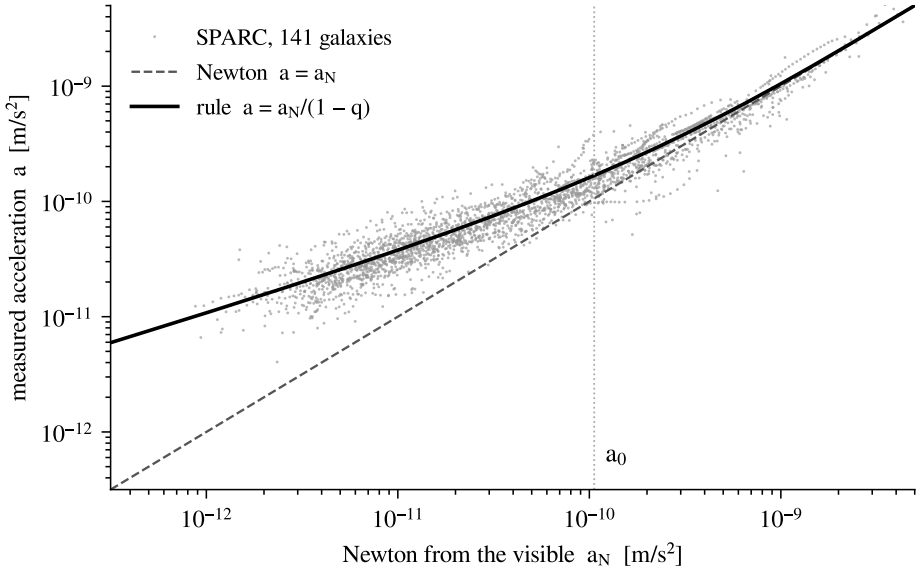


Figure 1. The rule at all measured points of the 141 galaxies (SPARC, catalogue values for distance and inclination, no fit per galaxy). Above the threshold a_0 the measured acceleration coincides with Newton. Below it the return holds it above what the visible alone would give.

Example 1: Atom and Nucleus

Hydrogen. The shell of hydrogen is a fast-orbiting vortex: it runs with $v = \alpha c$ on the Bohr radius a_B. Its binding is the energy of this orbit, $E = \frac{1}{2}\mu v^2$, with μ the reduced mass of electron and proton, because both orbit their common centre of mass:

$$E = \tfrac{1}{2} \cdot \mu \cdot (\alpha c)^2 = \mathbf{13.598\ eV}$$

Measured (PubChem): **13.598 eV**. No fitted number stands in between.

The nucleus. The nucleus is a held funnel. The length on which it holds its nucleons sits in the Coulomb term of the binding. Fitted to all nuclei of AME2020 (Figure 2), this term is $a_C = 0.69$ MeV, and from it

$$\lambda_H = (3/5)\cdot\alpha\hbar c / a_C = \mathbf{1.25\ fm} \qquad R(^{56}\mathrm{Fe}) = \lambda_H \cdot 56^{1/3} = \mathbf{4.8\ fm}$$

The same length comes by a second, independent route: mirror nuclei, in which protons and neutrons are exchanged, differ only in their winding. Their difference in binding gives **1.33 fm**. And the speed of the nucleons in the holder can be read twice: from the size, $v/c \approx \bar{\lambda}_p/\lambda_H = \mathbf{0.17}$, and from the depth of the binding of ^{62}Ni, **0.14**. Size and binding read the same.

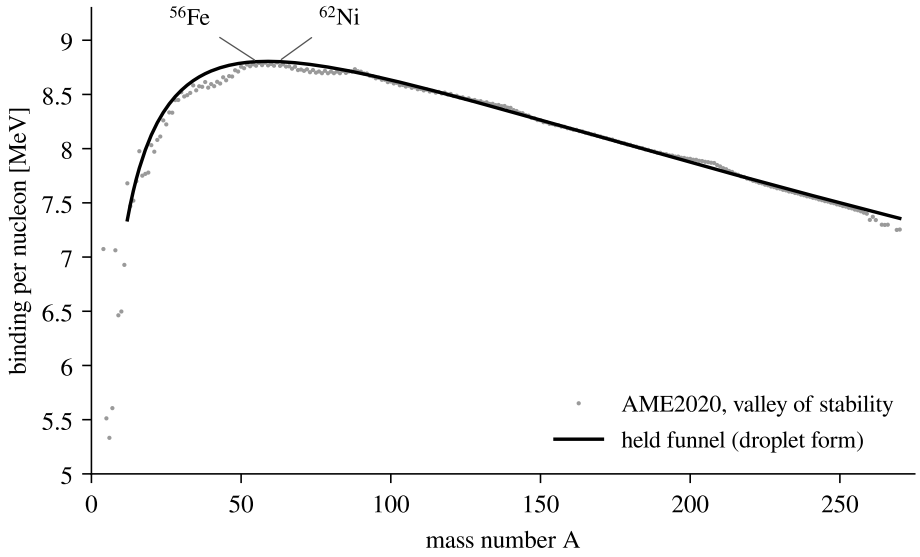


Figure 2. Binding per nucleon in the valley of stability (AME2020) and the held funnel in droplet form, whose Coulomb term reads the holder λ_H. The largest binding lies at ^{56}Fe and ^{62}Ni. There the burning of stars ends.

Example 2: The Solar System

The funnel of Earth and Sun, without return: all numbers follow from the fold $\Delta s = GM/(3c^2)$ and the clock rate $d\tau/dt = \sqrt{(1 - 6\Delta s/r)}$.

Body	Fold Δs	Limit radius $r_0 = 6\Delta s$	Clock at the surface, lag per year
Earth	1.478 mm	8.87 mm	0.022 s
Sun	492.2 m	2.95 km	67 s

The Earth follows the slope of the Sun's fold: $v = c\cdot\sqrt{(3\Delta s_\odot/r)}$ = **29.78 km/s**, one revolution in **365.3 days**. In the Sun's funnel its clock lags by **0.31 s** per year. Light at the solar limb is deflected by **1.7512″**, half by the clock rate and half by the extra length (0.8756″ + 0.8756″). The perihelion of Mercury advances by **42.98″** per century. The threshold of the Sun lies at r^* = **7480 AU**, far outside the planets: at the Earth's orbit the addition of the return is e^{-7480}, the solar system is Newton.

Example 3: The Closure Numbers

The shell fills by n + l and closes after 2, 10, 18, 36, 54, 86, 118. The periods are 2, 8, 8, 18, 18, 32, 32 long. The pure $1/r^2$ form would close after 2, 10, 28, 60. The held body closes after 2, 8, 20, 40, 70, 112. With the twist, which pulls down the state of largest winding in every shell, after O(k) + 2(k + 2) = 28, 50, 82, 126, and the next closure would be 184. None of these numbers is fitted, they are counted.

Measured (PubChem): the six strongest drops of the ionisation energy from one element to the next lie exactly after Z = 2, 10, 18, 36, 54, 86, from a factor **4.6** at He to **2.8** at Rn (Figure 3). Measured (AME2020): the six strongest jumps of the two-nucleon binding lie at N = 16, 20, 28, 50, 82, 126 and Z = 14, 20, 28, 38, 50, 82.

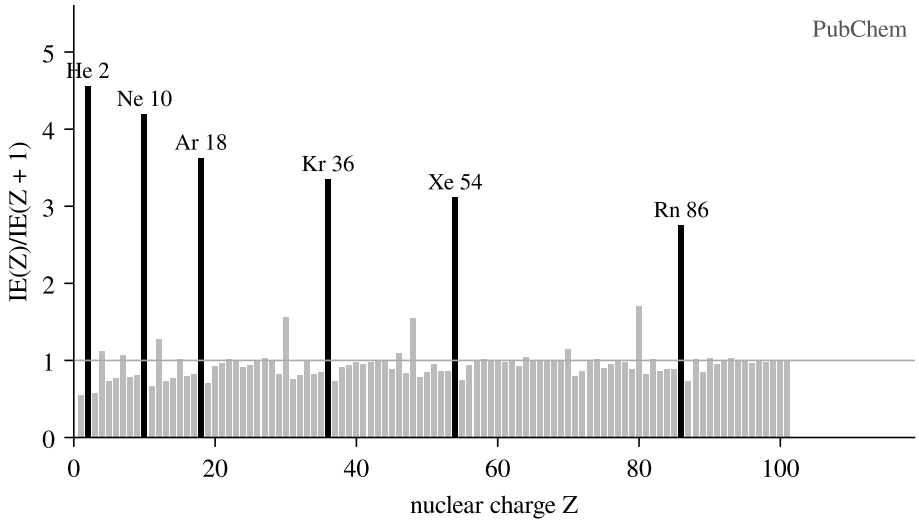


Figure 3. The drop of the ionisation energy IE(Z)/IE(Z + 1) over all elements (PubChem). The six strongest drops, in black, stand exactly at the closure numbers of the shell.

Example 4: Stars

Mass scale. $M_* = m_{Pl}^3/m_p^2$ = **1.853 M⊙**: the star as the mirror image of the Planck point at the quantum length of the proton.

Limiting mass. If the counting rule of the electrons holds the funnel, it has the form n = 3. Its equation $\theta'' + (2/\xi)\theta' + \theta^3 = 0$ is integrated from the centre: the edge lies at ξ_1 = **6.897**, the coefficient is $\omega_3 = -\xi_1^2\theta'(\xi_1)$ = **2.018**, and from it

$$M = \omega_3 \cdot (\sqrt{(3\pi)}/2) \cdot m_{Pl}^3/(2m_u)^2 = \mathbf{1.456\ M\odot}$$

Ceiling. The same funnel reaches at most $kT = \mathbf{0.096} \cdot (\mu/\mu_e) \cdot m_{ec}^2$, for carbon and oxygen (μ/μ_e = 7/8) $\mathbf{5.0 \cdot 10^8}$ **K**. The burning ladder ends where the binding per nucleon no longer grows, at x = ¼: in the valley at Z = **26.7**, between iron and nickel.

Carter. That the star on the mass scale lies at the border of radiation and convection requires $\alpha_G \approx \alpha^{12} \cdot (m_e/m_p)^4$: computed $\mathbf{2.0 \cdot 10^{-39}}$ against $\alpha_G = \mathbf{5.906 \cdot 10^{-39}}$, a factor **2.9**, of order one.

Example 5: The Fold

When the shell falls onto the centre, the funnel stays finite. A fold of 10^6 M⊙ in a galaxy has $s^2 = \mathbf{1.4 \cdot 10^{17}}$ at the horizon: it lies deep in the Newtonian regime, the return acts only outside r* = **36 pc**. The fold of one solar mass has the temperature $kT = \hbar c^2/(4\pi)$, $T = \mathbf{6.2 \cdot 10^{-8}}$ **K**, a period $\mathbf{2.3 \cdot 10^{22}}$ times shorter than the return ($kT_I = \hbar a_0/c$). Its entropy is $S = 4\pi k \cdot (M/m_{Pl})^2 = \mathbf{1.05 \cdot 10^{77}}$ k. If it radiates as a black body at its temperature, it lives $t = 5120\pi \cdot \alpha_G \cdot 3\Delta s/c = \mathbf{2.1 \cdot 10^{67}}$ years. What falls into it releases, down to the innermost stable orbit, **5.7 %** of its rest energy, **6.1** times as much as nuclear binding up to iron (^{56}Fe: **0.94 %**, AME2020). At the meeting point of rotation, j = 1, it is **42 %**. Only at $M_U/4 = \mathbf{1.4 \cdot 10^{23}}$ **M⊙** do horizon, threshold and closure length coincide.

Example 6: A Galaxy

NGC 2403 is a spiral galaxy with a well-measured rotation curve (SPARC, 73 points, quality 1). The calculation takes only the visible: gas, stellar disc and bulge as SPARC gives them from light and the 21-cm line, with the mass-to-light ratio of the disc $Y_d = 0.67$ and of the bulge $Y_b = 0.7$, the same for all galaxies. Y_d is not a quantity of the model but the conversion of light into fold. It follows from the demand that stars and gas obey the same rule. Distance and inclination stay at the catalogue values. Nothing is fitted to this galaxy.

Step by step at each radius:

1. $v_{bar}^2 = V_{gas} \cdot |V_{gas}| + Y_d \cdot V_{disc}^2 + Y_b \cdot V_{bulge}^2$ (the Newtonian speed of the visible)
2. $a_N = v_{bar}^2 / R$
3. $a = a_N / (1 - e^{-\sqrt{(a_N/a_0)}})$, $v = \sqrt{(a \cdot R)}$

R [kpc]	v_{bar} [km/s]	a_N/a_0	v rule [km/s]	v measured [km/s]
0.96	61	1.20	75	75 ± 1.0
2.98	85	0.74	112	100 ± 2.7
5.96	81	0.33	122	125 ± 0.9
9.94	71	0.16	124	127 ± 2.6
14.91	64	0.08	128	134 ± 0.6
19.87	55	0.05	125	136 ± 5.6

The visible alone falls off outwards, from 85 to 55 km/s. The rule keeps the curve flat, as measured (Figure 4). Over all 73 points the rule lies on average at 1.01 times the measurement, with a scatter of 0.03 dex.

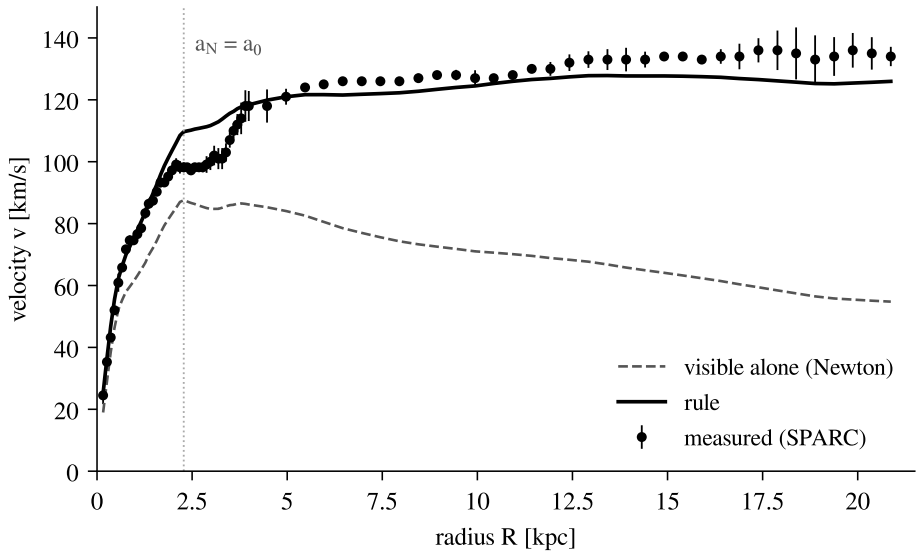


Figure 4. NGC 2403: the measured rotation curve (SPARC), the visible alone and the rule. Inside the dotted line a_N lies above a_0, there the rule is Newton. Outside, the return fraction grows and keeps the curve flat.

Not every galaxy hits this closely with the catalogue values: NGC 3198, computed the same way, lies on average at 1.16 times. Distance and inclination are measurements with their own error. Over all 141 galaxies they are therefore estimated galaxy by galaxy, and only this full calculation reads the threshold (Example 8).

Example 7: The Milky Way

As a point mass of $6 \cdot 10^{10}$ M⊙ the Milky Way has the fold Δs = **197.4 AU** and the threshold radius r* = **8.88 kpc**. There ν = **1.582**. Deep outside the curve becomes flat, $v^4 = GMa_0$: **170.4 km/s**.

R [kpc]	v Newton [km/s]	v rule [km/s]
5	227	249
8.88 (r*)	170	214
15	131	196
50	72	178

At 50 kpc the enhancement is ν = **6.14** (Figure 5). A uniform external field does not change it, the Milky Way falls with it. Far outside, the deflection of light tends to $\alpha_\infty = 2\pi \cdot (v_{flat}/c)^2$ = **0.42″**. The seam, where the local q reaches the cosmic one (a_N = **0.029**·a_0), lies at R_Q = **52.0 kpc**. No region around it binds beyond $R_{max} = v_{flat}/H_\infty$ = **3.07 Mpc**, the mean of threshold and closure, with $H_\infty = c \cdot \sqrt{\Lambda_s}$, the rate of the end state.

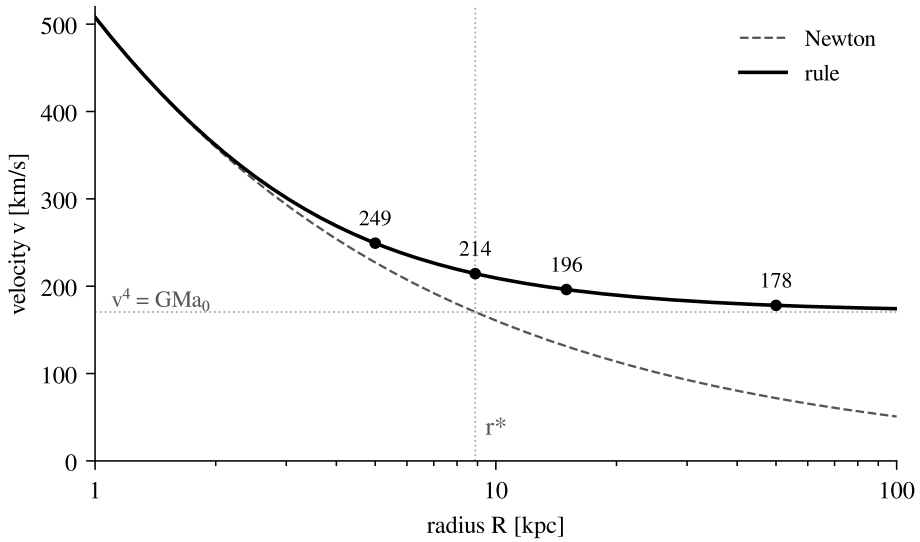


Figure 5. The Milky Way as a point mass: Newton falls off outwards, the rule bends around r* and runs flat towards $v^4 = GMa_0$.

Example 8: The Threshold from 141 Galaxies

Over all **141** galaxies (SPARC, quality 1 or 2, inclination above 30°) with **2688** points, distance, inclination and mass-to-light estimated for each galaxy: the mass-to-light ratio of the discs follows from the demand that stars and gas obey the same rule, Y_d = **0.67**. Then the threshold reads a_0 = **1.11**·10^{-10} m/s² (16 to 84 %: 1.00 to 1.21). The Bose form fits best: the simple form is worse by $\Delta\chi^2$ = **53**, the standard form by **41**. The return has dimension **0.92 ± 0.05**, one-dimensional as the square root in q requires.

Discs at the threshold. The disc scale length is, in the median, **0.71**·r* (16 to 84 %: 0.37 to 1.28), that is r*/√2, and it grows with the baryonic mass as $M_b^{0.31 \pm 0.02}$.

Single funnel or collective. Compact globular clusters (g = 25·a_0) show hardly any addition, ν = **1.007**. Thin systems (g = 0.13·a_0) the full one, ν = **3.3**. Pressure-supported spheres have $\sigma_v^4 = (4/81) \cdot GMa_0$: **9.1 km/s** at 10^7 M⊙, **19.3 km/s** at $2 \cdot 10^8$ M⊙. A whole system lies below the threshold at $\Sigma < a_0/(\pi G)$ = **242 M⊙/pc²**.

On the 52 star clusters of the Local Volume Database with measured dispersion the thin ones ($g_N < a_0$) lie +0.10 ± 0.03 dex above the prediction with the rule, +0.41 with Newton (χ^2 53 against 235).

Example 9: The Cosmos

From a_0 alone follows dark energy, $\rho_{DE} = 3\pi^3 a_0^2/(30Gc^2)$: each of the three loops of the torus carries the restoring of the fold. With ω_b, ω_c and T_{CMB}, the closed balance gives

$$H_0 = \mathbf{67.2}\ \text{km/s/Mpc} \qquad \Omega_{DE} = \mathbf{0.685} \qquad \Omega_m = \mathbf{0.315}$$

This can be tested on a quantity that depends on no calibration: the ratio of transverse to radial distance, D_M/D_H, which DESI measures from the baryon oscillations. In it, the length of the oscillation r_d and H_0 cancel. It reads only the shape of the expansion. The model has no free number here.

$$D_M/D_H(z) = E(z) \cdot \int_0^z dz'/E(z') \qquad E(z) = \sqrt{(\Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_{DE})}$$

z	DESI DR2	Model
0.51	0.621 ± 0.017	0.594
0.706	0.892 ± 0.021	0.878
0.934	1.223 ± 0.019	1.252
1.321	1.947 ± 0.045	1.997
1.484	2.381 ± 0.136	2.351
2.33	4.517 ± 0.097	4.548

Six values, $\chi^2 = 6.8$: on average the model lies one measurement error from the data, as a correct prediction should (Figure 6). On all precision data together (background radiation, lensing, DESI, Pantheon+, growth, light nuclei) the whole reads the dark energy today, form part and state part together, as $a_0 = 1.085 \pm 0.003$ when the maximum of the state part lies at the saturation of the loop, with as many free numbers as the standard model and $\Delta\chi^2 = 8.7$ below it.

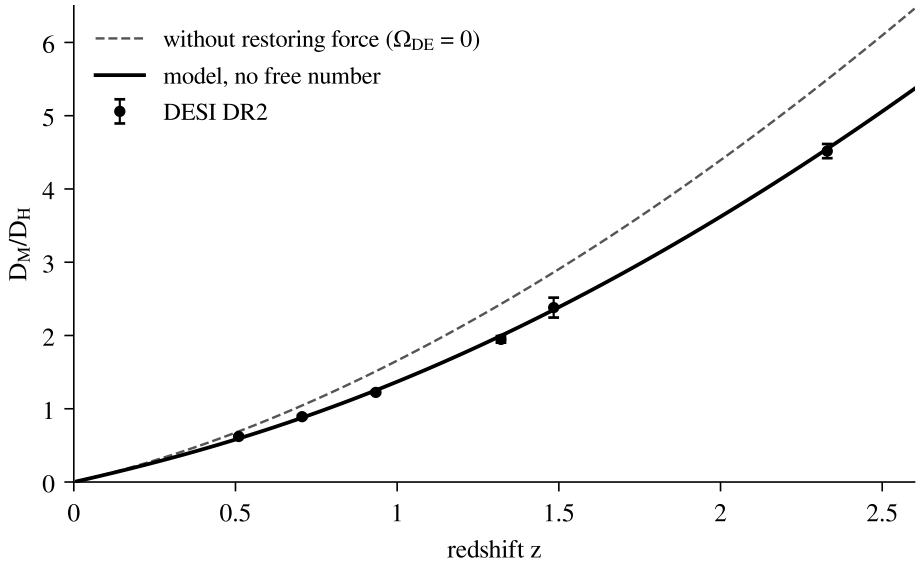


Figure 6. Transverse to radial distance, D_M/D_H, at DESI DR2 and the model without a free number. Dashed: the expansion without restoring force ($\Omega_{DE} = 0$), which runs too far at large z. The restoring of the three loops sets the shape.

Example 10: The Ladder from a_0

From the threshold alone follow the closure length $\Lambda = c^2/(2a_0)$ = **13.7 Gpc** and the return time $\tau_D = 2\Lambda/c$ = **89.6 billion years**, longer than the age of the universe: the return is a stationary loop at the place. With the Planck length comes the ladder number $\varepsilon_L = \ell_P/\ell_0$ = $\mathbf{1.9{\cdot}10^{-62}}$ and with it the rungs of the ladder of lengths (Figure 7):

Rung	**Mass**	**where two lengths are equal**
$M_{min} = \varepsilon_L{\cdot}m_{Pl}$	$4.1{\cdot}10^{-70}$ kg	$\bar{\lambda} = \ell_0$
$M_c = \varepsilon_L{}^{1/3}{\cdot}m_{Pl}$	$5.8{\cdot}10^{-29}$ kg	$r^* = \bar{\lambda}$
m_{Pl}	$2.2{\cdot}10^{-8}$ kg	$3\Delta s = \bar{\lambda} = \ell_P$
$M_U = m_{Pl}/\varepsilon_L$	$1.1{\cdot}10^{54}$ kg = $5.7{\cdot}10^{23}$ M⊙	$r^* = 3\Delta s = \ell_0$

The breathing: $\Theta = 2\pi a_0/(cH_0)$ = **1.020**, the vacuum part $\Omega_s = (\pi^2/45){\cdot}\Theta^2$ = **0.228**. Three loops give Ω_{DE}, today Ω_{DE}/Ω_s = **3.03** (read after Planck). The vacuum part is the geometric mean of the Planck density and the quantum density of the closure length: $\Lambda_s{\cdot}\ell_P{}^2 = (4\pi^4/15){\cdot}\varepsilon_L{}^2$ = $\mathbf{9.4{\cdot}10^{-123}}$. Its time-fixed number is $k_s = \pi^3/30$ = **1.034**. If it alone carried the expansion, this would run into $H_s = H_0{\cdot}\sqrt{\Omega_s}$ = **32.1 km/s/Mpc**, with three parts into **55.6 km/s/Mpc**. The whole lies with r^*/Λ = **0.971** just below its own meeting point. Its cell is L = **27.9 Gpc**.

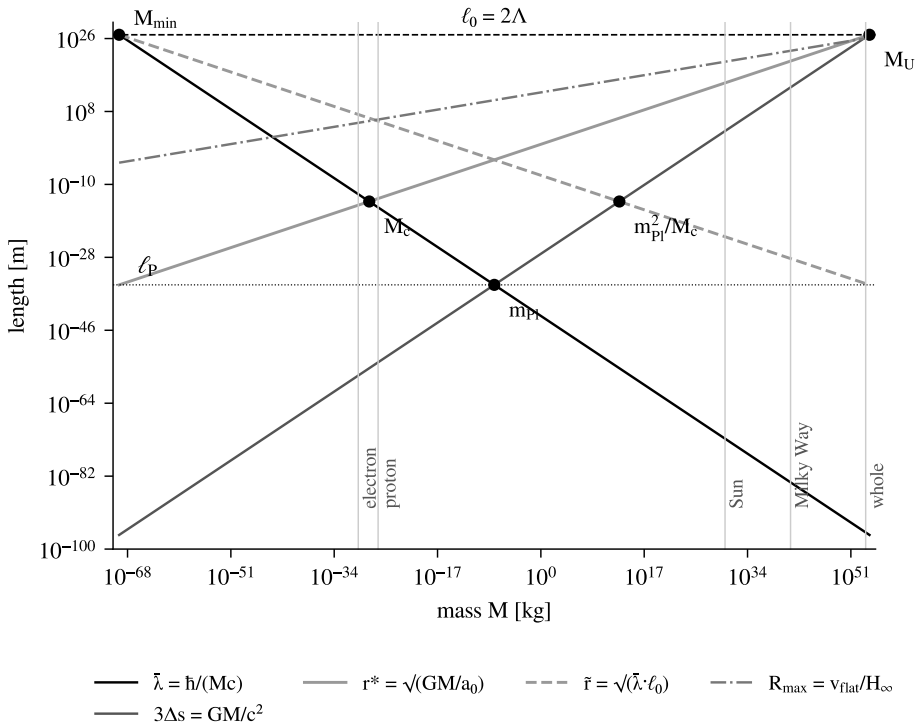


Figure 7. The ladder of lengths: quantum length $\bar{\lambda}$, fold $3\Delta s$, threshold radius r^* and return depth of the quantum motion $\tilde{r}$ over the mass, the largest turnaround $R_{max} = v_{flat}/H_\infty$, with the length of action ℓ_0 and the Planck length. Where two lengths become equal lies a rung, five meeting masses from M_{min} to M_U. The mirror $M \leftrightarrow m_{Pl}^2/M$ exchanges $\bar{\lambda}$ and $3\Delta s$ and with them r^* and $\tilde{r}$.

What the Ten Examples Show

From the atom to the whole, the same chain is used: fold, return, threshold. In the atom, in the nucleus, in the solar system and in the fold the accelerations lie far above the threshold, the return is zero, and fold and clock rate remain. In the galaxies the threshold is crossed, the curves stay flat, and 141 galaxies read it as one number. In the whole the same threshold is the restoring itself. It sets the ladder of lengths and the expansion. None of the ten examples needs a number chosen specifically for it.